Direct proof certificates and smaller Diophantine consistency equations

A materialized quartic in 3,614 natural unknowns and equations in 21, 20, and 19 integers of degrees 16, 18, and 40

Prepared for Vladimir Reshetnikov's ProveIt project

AI-assisted research contribution with reproducible exact algebra audits

24 October 2026

Repository snapshot: `08200eb10fde2db408c69dffd840d6920a569b1f`

Abstract

We give two complementary improvements to explicit Diophantine representations of ZFC refutability. First, we materialize a fixed quartic in 3,614 nonnegative integer unknowns, with 15,996 nonzero monomials and maximum absolute coefficient 127,778,545. Its construction compiles the repository's calibrated numerical proof calculus directly, without simulating a Turing machine or instantiating Jones's universal polynomial. A companion quartic has 6,183 unknowns and coefficient height 1,993. The central compilation step is a shared Chinese-remainder certificate: one global factorial makes every local range factorial invertible, allowing the latter to be removed and heterogeneous witness bounds to be used. The exported equations contain only ordinary polynomial monomials.

Second, integral triangular coordinate changes and two Pell-root translations turn an existing 25-integer octic into fixed equations in 21, 20, and 19 integers of exact degrees 16, 18, and 40. At these dimensions the best inspected incoming degrees were 20, 32, and 88. All new coordinate maps are polynomial bijections on zero sets. Composing the arguments with the explicitly identified inherited calibration theorems gives, in ZFC, equivalence between absence of solutions and Con(ZFC). This is an ordinary mathematical proof with a stated dependency boundary, not a new end-to-end proof-assistant certificate. Neither consistency nor global minimality is claimed.

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1 The precise result and the meaning of simplicity

Fix the usual arithmetized assertion Con(ZFC) that no finite ZFC proof of contradiction exists. The target is a fixed integer-coefficient polynomial P, with its domain specified, for which

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0\right) \iff \mathrm{Con(ZFC)}. \tag{1}$$

Here D is either $\mathbb{N} = \{0, 1, \ldots\}$ or $\mathbb{Z}$. These are different representation problems: a natural-number equation must not silently be advertised as an unrestricted integer equation. All coefficients are fixed; there is no free input encoding the theory.

There is no single established numerical meaning of the simplest Diophantine equation. Variable count, degree, coefficient height, number of monomials, circuit size, readability, and the complexity of the calibration proof compete. We therefore deliver several fully specified points, rather than conceal a large cost in an unspecified coefficient or arithmetic macro.

Equation and domain	Unknowns	Degree	Squared residuals	Coefficient information
P_{dir} over $\mathbb{N}$	3,614	4	2,954	height 127,778,545
P_{ht} over $\mathbb{N}$	6,183	4	5,523	height 1,993
Q_{21} over $\mathbb{Z}$	21	16	9	six supplied numerals
Q_{20} over $\mathbb{Z}$	20	18	8	same six numerals
Q_{19} over $\mathbb{Z}$	19	40	7	same six numerals

Table 1: Completed equations. Height means the maximum absolute coefficient of the expanded polynomial, not of a residual or a circuit.

Theorem 1.1 (Completed consistency representations)**.** *The five fixed polynomials in the table satisfy* (1) *on their stated domains, by the ordinary arithmetic arguments below composed with the repository calibration results specified in Section 9. Their variable counts and degrees are exact.*

The direct construction is useful when small coefficients, degree four, and a proof-calculus interpretation matter. The 19-integer construction is the most compact new equation here by variable count; it retains the large fixed numerical tuple of its inherited machine-based source. The two routes have different calibration dependencies.

1.1 What is new relative to the inspected repository

All comparisons refer to the snapshot `08200eb10fde2db408c69dffd840d6920a569b1f`, including the six archive roots in `docs/incoming`. The accompanying comparison inventory identifies each source. In particular, the incoming joint-guard package already contains a 25-integer octic and several small-variable projections. Reconstructing those equations is necessary calibration work, not a new result. The new coordinate projections improve the inspected $(n, \deg)$ points

$$(21, 20) \mapsto (21, 16), \quad (20, 32) \mapsto (20, 18), \quad (19, 88) \mapsto (19, 40).$$

No claim is made to improve existing nine-positive or ten-integer high-degree representations by variable count.

On the direct route, the repository already supplies a numerical Horn calculus, commutative proof certificates, and a bounded-factor carrier. Its large-carrier candidate reports 47,439,839 representation witnesses before a final ordinary-polynomial flattening. That number is a different compilation stage, so a ratio against our final 3,614 coordinates would not be a controlled like-for-like comparison. Our

contribution is the shared compilation theorem, the combination of concrete reductions, and a completed literal polynomial with small numerical coefficients. We found no such completed small-coefficient direct quartic among the inspected incoming packages. This is a repository comparison, not a claim of worldwide priority.

1.2 An unambiguous static definition of the direct polynomial

The file `direct/equation.json` lists 2,954 residuals. A residual is a finite list of pairs (c, μ), where $c \in \mathbb{Z}$ and μ is a list of zero, one, or two variable indices in $\{0, \dots, 3613\}$. Repetition of an index means a square. The empty list means the constant monomial. If the residual lists are $\mathcal{L}_0, \dots, \mathcal{L}_{2953}$, the single equation is, exactly,

$$P_{\mathrm{dir}}(X_0, \dots, X_{3613}) = \sum_{\ell=0}^{2953} \left(\sum_{(c,\mu)\in\mathcal{L}_\ell} c \prod_{j\in\mu} X_j \right)^2 = 0. \tag{2}$$

These lists are included as data, not left as the output of an unavailable compiler. The fully expanded coefficient list is also included in `direct/expanded_polynomial.json`. The builder reproduces both. The analogous two files in `direct/small_height` define P_{ht}. No exponentiation, binomial coefficient, factorial, division, bounded quantifier, or variable-length product remains in these files. Formula (2) is simply a compact convention for displaying a completely fixed polynomial.

2 The exact numerical calculus encoded by the source

The local source uses the numerical calculus C_{222}, independently of any machine that enumerates it. Its primitive codes are the following ordinary nonnegative polynomials:

$$\begin{aligned} I(a,b) &= 2(a+b)(a+b+1)+4b, \\ M(a,b) &= I(a,b)+1, & A(a,b) &= I(a,b)+2, & E(a,b) &= I(a,b)+3, \\ N(a) &= I(a,1), & X(a,b) &= N(A(a,N(b))). \end{aligned}$$

They are, respectively, the repository's implication, membership, universal quantification, equality, negation, and existential-quantification codes. The falsity code is 1, and the free truth code is 0.

The eight unconditional axiom codes are

$$\begin{aligned} r_1(a,b) &= X(a,E(a,b)), \\ r_2(a,b,c) &= I(A(a,I(b,c)),I(A(a,b),A(a,c))), \\ r_3(a,b,c) &= I(E(a,b),I(M(c,a),M(c,b))), \\ r_4(a,b,c) &= I(A(a,A(b,c)),A(b,A(a,c))), \\ r_5(a,b,c) &= I(E(c,a),I(M(c,b),M(a,b))), \\ r_6(a,b) &= I(X(a,A(a,b)),b), \\ r_G(a,b,c) &= I(E(c,a),I(E(c,b),A(M(a,b),E(a,b)))), \\ r_8(a,b,c,d) &= I(I(I(c,d),a),I(I(a,c),I(b,c))). \end{aligned}$$

The guarded seed bundle is also a fixed composition of these polynomials. To state it without a long

nested line, define

$$\begin{aligned}
R(b,k,h) &= A(b, I(h, N(A(b, I(h, N(M(b,k)))))))),\\
S(a,b,k) &= A(k, I(M(k,a), M(k,b))),\\
C_0(a,b,k,h) &= I(S(a,b,k), R(b,k,h)),\\
C_1(a,b,k,h) &= N(A(a, C_0(a,b,k,h))),\\
C_2(a,b,k,h) &= N(A(a, I(h, C_1(a,b,k,h)))),\\
C(a,b,k,h) &= A(b, I(M(b,k), C_2(a,b,k,h))),\\
V(a,b,c) &= I(M(a,b), A(c, M(a,b))),\\
K(a,b,c) &= A(c, I(C(a,b,c,c), N(R(b,c,c)))),\\
\mathcal{B}(a,b,c) &= N(I(V(a,b,c), K(a,b,c))).
\end{aligned}$$

The bundle is an axiom when $c \neq a$ and $c \neq b$. These conditions are represented using one natural witness d by

$$(c-a)^2(c-b)^2 - (d+1)^2 = 0.$$

Existence of such d is equivalent to the two disequalities: in the forward direction the product cannot be zero, and in the reverse direction take $d = |(c-a)(c-b)| - 1$.

There are precisely twelve branches in the padded two-premise relation: truth $(p,q,v) = (0,0,0)$; each of the eight unconditional axiom instances with $p = q = 0$; the guarded bundle with $p = q = 0$; modus ponens $q = I(p,v)$; and generalization $q = 0$, $v = A(a,p)$. The four natural schema coordinates a, b, c, d are shared across this finite disjunction. Their values need satisfy only the chosen branch.

An exact arithmetic audit, beyond random replay. The supplementary script `audit_calculus.py` independently implements the displayed formulas and compares them with the delivered source by sparse-polynomial arithmetic. A composite expression is replaced by a source wire name only after its coefficient dictionary equals that wire's defining polynomial exactly. In this way it recognizes all fifty-six branch-specific wire definitions and verifies all thirty-four remaining source residuals. Each replacement is a finite ring identity. This checks the arithmetic translation compositionally, without expanding the very large nested bundle polynomial and without treating a finite random sample as an identity proof.

The direct comparison is against `Code254.lean` and `KCode222.lean` at the pinned repository commit. The semantic dependencies are then

$$C_{222}(1) \iff \mathrm{BProv}(\mathrm{ZF}, \bot) \iff \mathrm{BProv}(\mathrm{ZFC}, \bot).$$

The named repository theorems are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. The latter uses the constructible-universe proof that ZF and ZFC are equiconsistent. The current construction inherits those mathematical calibration results. It does not claim to have rechecked their full Lean dependency closure, and the new compiler is not itself a Lean proof.

3 Why a finite commutative certificate is enough

This section records the mathematical content of the inherited direct flow mechanism [2], so that the bridge used by the compiler is visible. Let $\mathcal{H}$ be a finite list of instances of single-conclusion Horn rules with at most two premises. Axiom instances have zero premises. Pad missing premises with the freely available truth label 0, and let $e_0 = 0$. For a rule $(p,q) \to v$, assign the flow vector $e_v - e_p - e_q$ in the free abelian group on the nonzero labels.

Lemma 3.1 (Single-conclusion flow)**.** *A nonzero label b is derivable if and only if a finite multiset of rule instances has sum of flow vectors e_b.*

Proof. A finite proof can be expanded to a finite derivation tree. Every internal formula occurrence cancels between its production and its use; the root contributes e_b, and truth leaves contribute zero.

Conversely let D be the least set of labels containing truth and closed under the rules, and suppose $b \notin D$. Sum the coordinates outside D of each rule vector. A rule with conclusion outside D has at least one premise outside D, by closure, so its contribution is at most zero. A rule with conclusion in D also contributes at most zero. Every nonnegative integral sum of rule vectors therefore has nonpositive total outside D, whereas e_b has total one. This is a contradiction. Only the finitely many labels in the proposed certificate are needed for the finite sum. □

3.1 A carrier that avoids pairwise-prime formula weights

Take $t \geq 1$, $m \geq 1$, with every displayed formula label at most m, and define

$$M = (m+1)^t, \qquad w(a) = 1 + aM.$$

For t rule instances with conclusions v_i and premises p_i, q_i, consider

$$A(X) = \prod_{i<t}(1 + v_i X), \qquad C(X) = (1+X)\prod_{i<t}(1 + p_i X)(1 + q_i X).$$

Suppose $A(M) = C(M)$. First $\deg C \leq t$: otherwise $C(M) \geq M^{t+1}$, while

$$A(M) \leq (1 + mM)^t < ((m+1)M)^t = M^{t+1}.$$

Each of A and C now has at most t nonconstant linear factors, all with labels at most m. Its coefficient sum is at most $(m+1)^t = M$. Its constant coefficient is one, so every coefficient is strictly less than M. Uniqueness of ordinary base-M expansion implies $A = C$ in $\mathbb{Z}[X]$. Unique factorization over $\mathbb{Q}[X]$ then gives the flow equation with target 1: each positive label contributes its own distinct factor $1 + aX$. The free label zero contributes the unit. No coprimality of the numerical weights $1 + aM$ is assumed.

3.2 An accumulator with a polynomial local step

Introduce natural states $S_0, \ldots, S_t$ satisfying

$$S_0 = M + 1, \quad S_t = 1, \qquad S_{i+1}(1 + v_i M) = S_i(1 + p_i M)(1 + q_i M).$$

Every state is positive, by induction from $S_0 > 0$, since all weights are positive. Multiplying the local equations and cancelling the intermediate states gives $A(M) = C(M)$, hence a proof of the target.

Conversely expand a target derivation as a finite tree and traverse it by replacing a frontier conclusion by its premises. Let S_i be the product of the weights on that frontier. This supplies natural accumulator states satisfying every step, starting with $1 + M$ and ending with 1. Choose t to be the number of rule occurrences and m to bound the labels, then choose the carrier M by the displayed formula.

The states are encoded by one ordinary beta sequence

$$\beta(x, y, i) = x \bmod (1 + (i+1)y).$$

Choose y greater than every required state and divisible by $t!$. The moduli for $0 \leq i \leq t$ are pairwise coprime by the elementary gcd argument used below, so finite CRT supplies x with the prescribed digits. Crucially, this beta-sequence scale y, the carrier M, and the later shared-compiler scale T are chosen separately. Identifying these scales would introduce an unjustified size cycle.

4 A shared CRT compiler with binomial bounds

All variables in this section range over $\mathbb{N} = \{0, 1, 2, \ldots\}$. Let $E(k, z, w)$ and $F(k, z, w)$ be fixed polynomials with nonnegative integer coefficients, where k is one index, z is a finite tuple of external parameters, and $w = (w_1, \ldots, w_d)$ is a finite witness tuple. The nonnegative coefficient hypothesis is used only for monotonicity. For a proposed vector of witness bounds $u = (u_1, \ldots, u_d)$, choose a natural V with $u_j \le V$ for every j, and put

$$H = E(N, z, u) + F(N, z, u) + N + V + 1, \qquad T = H!, \qquad B = \prod_{k=0}^{N-1} (1 + (k+1)T).$$

The empty product is 1. In particular $T \ge 1$, $B \ge 1$, and $H \ge u_j + 1$ for every coordinate.

Theorem 4.1 (Shared binomial certificate)**.** *For fixed N, z, u, V as above, the following assertions are equivalent:*

1. *For every $k < N$ there is $w \in \mathbb{N}^d$ with $w_j \le u_j$ for all j and $E(k, z, w) = F(k, z, w)$.*
2. *There are $C, A_1, \ldots, A_d \in \mathbb{N}$ such that*

$$B \mid \binom{A_j}{u_j + 1} \qquad (1 \le j \le d), \tag{3}$$

$$B \mid 1 + (C+1)T, \tag{4}$$

$$E(C, z, A) \equiv F(C, z, A) \pmod{B}. \tag{5}$$

Proof. Write $D_k = 1 + (k+1)T$. We have $\gcd(D_k, T) = 1$. The numbers $D_0, \ldots, D_{N-1}$ are pairwise coprime: if $i < j < N$ and $g = \gcd(D_i, D_j)$, then $g \mid (j-i)T$ and $\gcd(g, T) = 1$, so $g \mid j - i$. But $j - i \le N \le H$, hence $j - i \mid H! = T$, and $g = 1$. Consequently $\gcd(B, T) = 1$ as well.

Suppose first that the arithmetic certificate exists. Fix $k < N$. Choose a prime $p \mid D_k$; this is possible since $D_k > 1$. Then $p \mid B$ and $p \nmid T$. A prime at most H divides $H!$, so $p > H$. For every j, the integer identity

$$(u_j + 1)! \binom{A_j}{u_j + 1} = \prod_{r=0}^{u_j} (A_j - r)$$

and (3) show that p divides the product on the right. This identity also holds when $A_j \le u_j$, because both sides are zero. Primality therefore supplies $w_j \in \{0, \ldots, u_j\}$ with $A_j \equiv w_j \pmod{p}$. Comparing (4) with $D_k \equiv 0 \pmod{p}$, and cancelling T modulo p, gives $C \equiv k \pmod{p}$. Polynomial evaluation and (5) imply

$$E(k, z, w) \equiv F(k, z, w) \pmod{p}.$$

Monotonicity gives

$$0 \le E(k, z, w) \le E(N, z, u) < p, \qquad 0 \le F(k, z, w) \le F(N, z, u) < p.$$

These congruent natural numbers are equal, proving the bounded row.

Conversely, choose one bounded witness $w^{(k)}$ for each $k < N$. By the Chinese remainder theorem and the pairwise coprimality just proved, choose natural C, A_j satisfying

$$C \equiv k \pmod{D_k}, \qquad A_j \equiv w_j^{(k)} \pmod{D_k} \quad (k < N,\ 1 \le j \le d).$$

The latter congruence makes the falling-factorial product vanish modulo D_k, since its factor indexed by $w_j^{(k)}$ vanishes. Thus

$$B \mid (u_j + 1)! \binom{A_j}{u_j + 1}.$$

Now $(u_j+1)!$ divides $H! = T$ and $\gcd(B,T) = 1$; we may cancel this factor to obtain (3). The index congruences give (4); the valid rows and polynomial congruence give (5). Pairwise coprimality combines the congruences modulo the D_k into congruences modulo B. When $N = 0$, there are no row obligations and $B = 1$, so both directions hold without any separate side condition. □

Why the factorial cancellation matters. The repository's supplied CRT certificate uses $B \mid (u_j+1)!\binom{A_j}{u_j+1}$ separately for every coordinate. Its exact source does not require $H \geq u_j + 1$, so simply dropping these factorials from that source would be unjustified. The explicit added bound $V + 1$ repairs this issue. It also permits heterogeneous bounds at no additional factorial cost. A representation of each binomial coefficient is still necessary; the theorem does not eliminate that obligation.

4.1 A finite arithmetic expression for the modulus product

The product defining B is explanatory notation and is eliminated before the final polynomial is issued. Set

$$Q = (1 + (N+1)T)^N + 2, \qquad L = TQ - 1.$$

Then the exact natural-number identity is

$$B = \left(T^N N! \binom{Q+N}{N} \right) \bmod L. \tag{6}$$

Indeed, the numerator equals $\prod_{i=1}^{N}(TQ + iT)$ and reduces modulo L to $\prod_{i=1}^{N}(1 + iT) = B$. Further,

$$B \leq (1+NT)^N < (1+(N+1)T)^N + 1 \leq TQ - 1 = L,$$

with the strict inequality also valid for $N = 0$, when the left side is 1 and the middle upper bound is 2. Thus B is the remainder. The denominator is positive because $T \geq 1$ and $Q \geq 3$. This is the same explicit product mechanism used by the repository's `BoundedForallRepresentation.productGraph`; the elementary calculation above records the needed identity independently.

Powers, binomial coefficients, factorials, and the final remainder are each replaced by their fixed finite polynomial graph. In particular, no variable-length product, variable exponent, or bounded universal quantifier remains in the emitted ordinary polynomial.

4.2 Keeping the majorants as arithmetic circuits

A signed arithmetic circuit P admits natural polynomial circuits E, F such that $P = E - F$. The recursive pairs are $(X, 0)$ for a variable, $(c, 0)$ or $(0, -c)$ for an integer constant according to its sign,

$$(E_1, F_1) + (E_2, F_2) = (E_1 + E_2, F_1 + F_2),$$

$$(E_1, F_1) - (E_2, F_2) = (E_1 + F_2, F_1 + E_2),$$

and

$$(E_1, F_1)(E_2, F_2) = (E_1E_2 + F_1F_2,\ E_1F_2 + F_1E_2).$$

Every operation preserves nonnegative coefficients. Repeated subexpressions are shared as nodes in a finite directed acyclic graph. Expanding this DAG into a tree is mathematically harmless but can make the representation far larger. The compiler uses node values as natural witnesses and emits their quadratic defining equations only once per use-site evaluation.

5 Eliminating local Horn wires before bounded compilation

The pinned `affine_shared_step.json` source contains twelve Horn branches. Its existential coordinates include thirteen shared names $q_0, \dots, q_{12}$. In each branch the actual wire definitions are triangular polynomials with nonnegative coefficients, and the first rows of that branch assert exactly $q_i - f_i = 0$. Consequently, on natural inputs, the definitions determine a unique natural value for every wire used by that branch. Substitution removes all thirteen existential wire coordinates.

There are fifty-six branch-specific definition occurrences and thirty-four remaining branch residuals. These are verified from the pinned JSON by exact sparse-polynomial comparison. Branch definitions use separate scopes. This does not assume one set of wire values makes all branches vanish: existential quantification distributes over the finite disjunction, and precisely one vanishing branch is sufficient. The product of the twelve branch sums of squares implements that disjunction.

The affine step originally has twenty-three natural witness coordinates. Wire elimination leaves ten:

$$p, q, v, a, b, c, d, \delta_p, \delta_q, \delta_v.$$

Here $(p, q) \to v$ is the rule, a, b, c, d are its four schema witnesses, and the three deltas implement $p, q, v \le m$. Using Theorem 4.1 with heterogeneous bounds removes the latter three witnesses too: bound p, q, v directly by m in the CRT certificate. Bound all remaining per-row witnesses by $V = m + U$, where U is one additional outer natural variable. Such U always exists for any finite valid certificate.

5.1 Thirteen per-row coordinates

Retain the five outer parameters t, x, y, m, M_0 and put $M = M_0 + 1$. The carrier condition is $M = (m+1)^t$. At row $i < t$, set $D_i = 1 + (i+1)y$ as an arithmetic expression, without introducing a separately bounded row coordinate. Besides the seven Horn coordinates p, q, v, a, b, c, d, use six row coordinates S, S', u, r, u', r'. The four beta equations are

$$\begin{aligned} x - S - D_i u &= 0, & S + r + 1 - D_i &= 0, \\ x - S' - (D_i + y)u' &= 0, & S' + r' + 1 - (D_i + y) &= 0. \end{aligned}$$

Since $D_i, D_i + y \ge 1$, these say exactly that S and S' are the consecutive beta digits. The remaining accumulator equation is

$$S'(1 + vM) = S(1 + pM)(1 + qM).$$

Let P be the product of the twelve substituted Horn branch sums of squares, plus the square of this accumulator residual and the squares of the four beta residuals. It is a fixed nonnegative polynomial in the index, the external parameters and these thirteen row coordinates. Split its arithmetic circuit as $P = E - F$ as above.

For the first three encoded coordinates use the bound m, and for the other ten use $V = m + U$. Form

$$H = E(t, z, u) + F(t, z, u) + t + V + 1, \qquad u = (m, m, m, V, \dots, V),$$

where $z = (t, x, y, m, M_0)$ denotes the external parameters (the polynomial need not actually use t except through the index bound). The shared certificate then expresses all rows with thirteen binomial graphs and no per-coordinate factorial graphs.

The endpoints need four natural witnesses u_0, r_0, u_t, r_t and the rows

$$\begin{aligned} x - (M_0 + 2) - (y + 1)u_0 &= 0, \\ M_0 + r_0 + 2 - y &= 0, \\ x - 1 - (1 + (t+1)y)u_t &= 0, \\ r_t + 2 - (1 + (t+1)y) &= 0. \end{aligned}$$

They assert $\beta(0) = M + 1$ and $\beta(t) = 1$ exactly.

Logical dependency boundary. The changes in this section preserve the numerical C222 Horn rule predicate of package 68. They do not supply a new proof that this predicate is equivalent to ZFC refutability. That calibration is inherited from the repository's pinned C222-to-ZF and constructible-choice arguments, including the sealed target code 1 and free truth code 0. The equivalence of complete accumulator certificates with finite Horn proofs is the pinned single-conclusion flow theorem, composed with the bounded-factor carrier theorem. Given those dependencies, all subsequent transformations here are explicit arithmetic equivalences provable in ZFC; none assumes Con(ZFC).

The pertinent sources are pinned to the repository snapshot on the title page:

- The DiophantineConsistency Lean modules for bounded-forall certificates and their representation, Pell equations, exponentiation, and factorials.
- Package 68's `COMMUTATIVE_CERTIFICATES.md` and `affine_shared_step.json`.
- The candidate `direct-flow-large-carrier/README.md`.

The provenance inventory supplies their complete repository paths and the immediate source copies.

6 Specializing every power graph to positive arguments

The completed source can choose $t \geq 1$ and $m \geq 1$ without losing a certificate. A zero-length accumulator path contradicts its distinct endpoints, and if all labels were zero then every accumulator update would leave the state unchanged. Hence the existence predicate is preserved by the parameterizations $t = t_0 + 1$, $m = m_0 + 1$.

Likewise each encoded CRT coordinate may be required positive. The Chinese remainder theorem specifies residue classes only; adding a positive multiple of B preserves all required congruences. Thus one may parameterize $A_j = \alpha_j + 1$ for natural α_j. This affects encoded coordinates, not the decoded row witnesses: the latter may still be zero, including the free truth code. The range certificate remains valid because its proof already establishes divisibility of the binomial value for every CRT representative of the prescribed small residues.

Under these choices every power graph used in the construction has positive base and positive exponent. The carrier and modulus powers have exponent $t > 0$; the factorial graphs are called at t and H, both positive; their radius and auxiliary binomial arguments are positive. In each coordinate-range binomial graph the upper argument A_j is positive and the lower argument $u_j + 1$ is positive. Therefore its three power calls 2^{A_j}, $(2^{A_j} + 2)^{A_j}$, and $(2^{A_j} + 1)^{u_j+1}$ are positive cases. The zero-base and zero-exponent branches of a general power graph are unnecessary in this specialized construction. Their removal is justified by this argument inventory; it is not a correct optimization of the unrestricted exponentiation relation.

For the positive-power certificate, its final Pell norm gives

$$A^2 = 1 + W^3(W + 2)Z^2, \qquad W \geq n \geq 1, \quad A \geq 2.$$

Consequently $Z \geq 1$ and $A > W \geq n$. Hence the natural subtraction $A \dot{-} n$ in the original power theorem equals ordinary integer subtraction. The congruence can be written with nonnegative polynomial sides as

$$X + Yn + Tq_+ = YA + m + Tq_-.$$

This is the only subtraction-elimination point that needs a separate inequality argument in the power specialization.

7 Eliminating every arithmetic macro

This section specifies the fixed finite graph certificates used by the direct materialization. All coordinates here are nonnegative integers. A displayed product or square is subsequently expressed by addition and multiplication gates; the final gate rows are quadratic.

7.1 A positive-power graph with 27 auxiliary coordinates

For positive inputs n, k and proposed output m, write

$$\mathcal{N}(a, x, y) = x^2 + y^2 - 1 - a^2 y^2.$$

The positive-power graph has the following 27 additional coordinates:

$$W, A, T, Z, X, Y, U, V, S, R, B,\ a_0, y_0, b_0, q, d,$$
$$b_+, b_-, s_+, s_-, r_+, r_-, x_+, x_-, g, w_n, w_k.$$

Impose the finite equations

$$A = a_0 + 2, \qquad Y = k + y_0, \qquad B = b_0 + 2, \tag{7}$$
$$\mathcal{N}(A, X, Y) = 0, \qquad \mathcal{N}(A, U, V) = 0, \qquad \mathcal{N}(B, S, R) = 0, \tag{8}$$
$$B = 1 + 4Yq, \qquad V = Y^2(d + 1), \tag{9}$$
$$B + Ub_+ = A + Ub_-, \qquad S + Us_+ = X + Us_-, \qquad R + 4Yr_+ = k + 4Yr_-, \tag{10}$$
$$X + Yn + Tx_+ = YA + m + Tx_-, \tag{11}$$
$$2An = T + n^2 + 1, \qquad T = m + g + 1, \tag{12}$$
$$W = n + w_n, \qquad W = k + w_k, \qquad \mathcal{N}(W + 1, A, WZ) = 0. \tag{13}$$

Existence of these coordinates is equivalent to $m = n^k$. This is the positive specialization of the Pell–Matiyasevich power criterion [3, 4], with every strict inequality represented by a natural slack, every divisibility represented by a quotient, and every possibly signed congruence quotient represented as a difference of two naturals. The first three norms and congruences characterize the kth Pell solution at parameter A. The last norm, the bounds on W, the modulus definition, and the strict output bound are precisely the remaining power criterion.

The condition $Y \geq k \geq 1$ excludes the exceptional zero-index Pell solution. The modulus $4Y$ is positive. Also (13) gives

$$A^2 = 1 + W^3(W + 2)Z^2.$$

Since $A \geq 2$ and $W \geq n \geq 1$, one has $Z \geq 1$ and $A > W \geq n$. Thus the integer rearrangement in (11) is equivalent to the original natural-subtraction congruence involving $A - n$. These side conditions are part of the proof, not optional simplifications.

The optional unrestricted builder retains the zero-exponent and zero-base branches. The delivered constructions do not need those branches, because the preceding section and an independently checked call-site induction establish positive inputs to all 55 power uses.

7.2 Binomial coefficients by a base digit

For $n, k \in \mathbb{N}$, put

$$q = 2^n + 1, \qquad L = q^k, \qquad P = (q + 1)^n.$$

The binomial theorem expands P in base q, with the coefficient $\binom{n}{k}$ in digit position k, because all coefficients are at most $2^n < q$. In particular, positions above n have digit zero. For proposed output c, add just four natural coordinates h, r, r_s, c_s besides the three power graphs, and impose

$$P = L(qh + c) + r, \qquad L = r + r_s + 1, \qquad q = c + c_s + 1. \tag{14}$$

The last two equations say $0 \le r < L$ and $0 \le c < q$. Uniqueness of quotient, remainder, and base digit proves that (14) holds exactly when $c = \binom{n}{k}$.

7.3 Factorials as a ceiling quotient

For $n \in \mathbb{N}$, define

$$\rho = (n+2)^{n+2}, \qquad c = \binom{\rho + n}{n}, \qquad P = (\rho + 1)^n.$$

Then

$$n! = \left\lceil \frac{P}{c} \right\rceil. \tag{15}$$

For $n = 0, 1$ this is immediate. For $n \ge 2$,

$$\frac{P}{c} = n! \prod_{i=1}^{n} \frac{\rho + 1}{\rho + i} = n! \prod_{i=1}^{n} \left(1 - \frac{i-1}{\rho + i}\right).$$

Using $1 - \prod(1 - a_i) \le \sum a_i$ for $0 \le a_i \le 1$, we obtain

$$0 \le n! - \frac{P}{c} < \frac{n!\, n(n-1)}{2(\rho + 1)} < 1.$$

The final strict bound follows, for example, from $n!\, n(n-1)/2 < n^{n+2} < (n+2)^{n+2} = \rho$. This proves (15), including the case where the quotient is already integral.

For proposed output f and two natural coordinates r, s, impose

$$P + c = cf + r + 1, \qquad c = r + s + 1.$$

Since $c > 0$, these say $c(f-1) < P \le cf$, which is equivalent to $f = \lceil P/c \rceil$. The required powers and binomial graph are those already specified.

7.4 Ordinary quadratic rows and one quartic

Every arithmetic-circuit addition or multiplication is represented by a new natural coordinate z and the row

$$z - x - y = 0 \qquad \text{or} \qquad z - xy = 0.$$

Numerals and shared nodes are treated once at each distinct evaluation. Signed equality is kept as an equality between two nonnegative circuits. Each graph obligation is then an ordinary residual of degree at most two. A finite conjunction of rows $R_\ell = 0$ is equivalent over $\mathbb{N}$ to the one equation $\sum_\ell R_\ell^2 = 0$.

Before affine elimination, the resulting fixed system has 6,183 coordinates and 5,523 rows. Every residual coefficient has absolute value at most four. Its expanded squared sum has 17,430 monomials and coefficient height 1,993. These are measured exact coefficient dictionaries, rather than an asymptotic bound.

7.5 Removing nonnegative affine gates

Suppose a retained gate has the defining row $z - L = 0$, where L is an affine polynomial with nonnegative integer coefficients in earlier coordinates. Substituting L everywhere and deleting this defining row and coordinate is a bijection on natural zero sets: the inverse restores $z = L$. Substitution of an affine polynomial preserves the quadratic degree bound.

We apply this operation in topological order when the expanded affine replacement has at most four terms. The support threshold controls coefficient and expression growth; it is unrelated to correctness. Exactly 2,569 coordinates and rows disappear. This yields

$$6183 - 2569 = 3614, \qquad 5523 - 2569 = 2954.$$

The resulting quartic has 15,996 monomials and coefficient height 127,778,545, which is less than 2^{27}. The increase in coefficient height is the explicit cost of removing these affine wires.

Measured quantity	Retained wires	Affine elimination
Outer coordinates	5	5
Other witness coordinates	1,665	1,665
Gate coordinates	4,513	1,944
Power / binomial / factorial graphs	55 / 16 / 2	55 / 16 / 2
Quadratic rows	5,523	2,954
Expanded monomials	17,430	15,996
Maximum residual coefficient	4	4,556
Maximum quartic coefficient	1,993	127,778,545

Table 2: Exactly measured direct materializations of the same certificate.

8 Coordinated changes of variables: 21, 20, and 19 integers

The incoming *Joint Sign Guards* package already supplies a 25-integer octic and equations in 21, 20, and 19 integers of degrees 20, 32, and 88. Thus reproducing the octic is a calibration step, not a new frontier claim. The following coordinated changes of variables give new equations in the same respective dimensions of exact degrees **16, 18, and 40**. They use the same complete source-34 numerical tuple and the same two determinant scales. Every transformation below is an integer-polynomial bijection on zero sets after an integral triangular change of coordinates.

8.1 The fixed coefficients and the common abbreviations

The fixed coefficients are

$$\beta = 2^{1064921} - 1, \qquad K_0, M_C, M_F, \qquad A = 2^{1113}, \quad B = 2^{666104}.$$

Here K_0, M_C, M_F are the fully supplied source-34 integers; M_F is already shifted. The six numerals are bound by the packaged coefficient manifest and reconstructed from the literal machine/table recipe. They occupy 615883 minimal unsigned bytes. They are fixed coefficients, never existential parameters. In particular $\beta > 0$.

For the 21-variable presentation take

$$J_3, C, z, f, h, r, v, w_3, s, k, y, \gamma, o, j, Y, c, I, t, p_1, p_2, p_3 \quad \in \mathbb{Z}.$$

The letter s is the shifted first Pell root (`tau_gap` in the literal packets); it has no connection with the scale variable v. Define polynomial abbreviations

$$\begin{gathered}
J_0 = 3J_3 + 1, \qquad W = 4(3w_3 + 1), \qquad q = \beta J_0 + 1, \qquad X = Wq, \\
F = (K_0 + W)C - z(q-1), \qquad E = XY, \qquad a = (X+1)Y, \\
\eta = c - Yk, \qquad \zeta = k - \eta, \qquad V = fo - c, \\
\Delta = a^2 + 4a + 3, \qquad D = X + ac + \gamma(4a+3), \\
R = (q-F)(q^3-q) - C(q^2-1) + (M_C + qM_F)J_0, \\
G = AFBC - (q - AF - BC)^2, \qquad Z = \eta\zeta, \\
T = 1 + p_1^2 + p_2^2 + p_3(p_3+1).
\end{gathered}$$

The alignment is typographical only: all names on the left are polynomial abbreviations and add no unknowns.

The nine residuals of the 21-variable equation are

$$\begin{aligned}
R_1 &= Y - (2v+1)q^3, \\
R_2 &= s^2 + (2s-k)EYk - 1, \\
R_3 &= D^2 - \Delta c^2 - 1, \\
R_4 &= k - 1 - hE - R, \\
R_5 &= V - jc + k - 1 - hE, \\
R_6 &= I - (2r+1)c^2, \\
R_7 &= I^2 - \Delta(f^2 - 1), \\
R_8 &= y^2 - 1 - I^2(y^2 - V^2), \\
R_9 &= GZ - (t^2 - G - Z + 1)^2 - T.
\end{aligned}$$

Set $Q_{21} = \sum_{i=1}^{9} R_i^2$. The displayed degrees are, in this order,

$$4, 6, 8, 5, 4, 3, 8, 6, 8.$$

In the highest-degree part of $Z = (c - Yk)(k - c + Yk)$, the monomial $-Y^2k^2$ occurs with coefficient -1. The coefficient of Y^4k^4 in R_9 is consequently -1, and no other residual contributes Y^8k^8 to its square. Thus

$$[Y^8k^8]Q_{21} = 1,$$

and the exact degree is sixteen, including after all fixed ports are specialized to their actual values.

8.2 Why these coordinates are effective

The literal incoming 25-integer packet uses $k_0 = 1 - V + jc + hE$ in three arithmetic rows, while retaining $\eta + \zeta$ in its quotient and positivity rows. To obtain the convenient uniform presentation used here, put $\varepsilon = V - jc + \eta + \zeta - 1 - hE$. Add $(E - a)\varepsilon$ to its main-ordinate row, add $-E(c - \eta)\varepsilon$ to its first-norm row, and add ε to its packed-rank row. Keep ε itself. These are reversible elementary row operations, so the integer zero set is unchanged. The exact checker compares all thirteen resulting rows with the independently reconstructed packet.

In this presentation of the source 25-integer octic use the integral triangular changes

$$C = q - F - \alpha, \qquad Y = a - E, \qquad k = \eta + \zeta, \qquad s = \tau - E(c - \eta).$$

Their inverses are

$$\alpha = q - F - C, \quad a = E + Y, \quad \zeta = k - \eta, \quad \tau = s + E(c - \eta).$$

The source rows then determine four coordinates monically:

$$F = (K_0 + W)C - z(q - 1), \qquad E = XY, \qquad \eta = c - Yk, \qquad V = fo - c.$$

Delete the four defining rows and substitute these four integer polynomials everywhere. This reduces 25 coordinates and 13 residuals to 21 coordinates and 9 residuals. The first norm is controlled by the exact identity, with $u = E(c - \eta) = EYk$,

$$(s + u)^2 - (k + u)^2 + (k + u)k - 1 = s^2 + (2s - k)u - 1.$$

The cancellation prevents a needless squaring of the degree-five expression EYk.

Retaining both Y and k is essential to the bound. It keeps η, ζ quadratic, even after E and F have been removed. The determinant G and product Z are both quartic, so the combined positivity residual remains of degree eight. Expanding the quotient coordinates too early was responsible for the larger degree of the inspected incoming projections.

For complete rigor, map an old zero forward by the four triangular definitions and then forget F, E, η, V. The four deleted source equations imply precisely the restoration polynomials above. Conversely, restore these coordinates at any new zero and invert the triangular changes; every source residual vanishes, including the four deleted defining equations. The two maps are inverse on zero sets. There is no division, denominator condition, or sign assumption in this argument. Therefore the inherited equivalence with finite ZFC refutability transfers to Q_{21}.

8.3 Twenty integers and degree eighteen

In Q_{21} make the integral change $y = fo + b$, where b is unrestricted, and eliminate $I = (2r + 1)c^2$ using R_6. All other coordinates remain. The key cancellation is

$$(fo + b)^2 - (fo - c)^2 = (c + b)(2fo + b - c).$$

The resulting auxiliary residual is

$$(fo + b)^2 - 1 - (2r + 1)^2 c^4 (c + b)(2fo + b - c),$$

of degree nine. The seven other residuals have degrees at most eight. The final equation $Q_{20} = 0$ is their sum of squares, has twenty integer coordinates, and has exact degree eighteen because

$$[r^4 c^{10} f^2 o^2] Q_{20} = 64.$$

Only the auxiliary row contributes this monomial. The variable change and monic elimination are reversible over the integers. Hence Q_{20} has a zero exactly when Q_{21} does.

8.4 Nineteen integers and degree forty

Finally substitute

$$Y = (2v + 1)\big(\beta(3J_3 + 1) + 1\big)^3$$

in Q_{20} and remove R_1. This leaves the nineteen coordinates

$$J_3, C, z, f, h, r, v, w_3, s, k, b, \gamma, o, j, c, t, p_1, p_2, p_3.$$

Use exactly the common abbreviations above, the additional definitions $I = (2r+1)c^2$ and $y = fo + b$, and the seven residuals $R_2, R_3, R_4, R_5, R_7, R_8, R_9$. These are completely specified ordinary polynomials, with respective degrees

$$12, 14, 7, 7, 14, 9, 20.$$

Their squared sum is Q_{19}. Its exact degree is forty, since

$$[v^8 J_3^{24} k^8] Q_{19} = 2^8\, 3^{24}\, \beta^{24} > 0.$$

This coefficient is the image of the $Y^8 k^8$ certificate. Every other residual has degree at most fourteen before squaring, so it cannot alter the degree-forty coefficient. The monic restoration of Y again gives a bijection of integer zero sets.

8.5 Proof boundary and measured comparison

Each new system therefore has a zero exactly when the existing 25-integer octic has a zero. Combining with its named native/compiler, signed-Pell, determinant and K215/refutation calibration gives, by ordinary mathematics formalizable in ZFC,

$$\mathrm{ZFC} \;\vdash\quad \neg\exists \boldsymbol{x} \in \mathbb{Z}^n\, Q_n(\boldsymbol{x}) = 0 \quad\longleftrightarrow\quad \mathrm{Con(ZFC)}, \qquad n \in \{19, 20, 21\}.$$

No absence of zeros or consistency assertion is proved here. No new end-to-end proof-assistant certificate or internally encoded derivation in the repository's object-language ZFC calculus is claimed.

The comparison includes all six incoming archives visible at the pinned repository snapshot, not just numbered proposals. The changes in $(n, \deg)$ are

$$(21, 20) \longrightarrow (21, 16), \qquad (20, 32) \longrightarrow (20, 18), \qquad (19, 88) \longrightarrow (19, 40).$$

These are degree reductions of 20%, 43.75%, and approximately 54.55% at unchanged dimensions. Other measures may increase; fixed coefficient payload size is unchanged. Existing ten-integer and nine-positive high-degree equations remain different tradeoffs. No worldwide priority or global minimum is asserted.

Exact finite checks verify every defining-row identity, both integral root translations, all compact-display residuals, all variable counts and liveness, the stated residual degree bounds, and the three nonzero coefficient certificates. They also reconstruct the complete numerical tuple. These computations audit the literal algebra; the zero-set maps above supply the universal proof.

9 Logical scope, semantic dependencies, and exact audits

9.1 Composing the direct proof

The direct direction from a hypothetical zero to a refutation is:

1. Restore the eliminated nonnegative affine gates. A sum of squares vanishes precisely when every residual does.
2. Read each arithmetic gate and each positive-power, binomial, factorial, and remainder certificate. Their graph theorems give their intended exact natural values.
3. Apply the shared binomial certificate to obtain all t valid bounded local rows. The beta equations and endpoint equations produce one complete accumulator path.
4. Apply the carrier and single-conclusion flow arguments to obtain a finite C_{222} derivation of the sealed target 1.

5. Apply the pinned C222-to-ZF and constructible-choice calibrations to obtain a finite ZFC refutation.

Every implication reverses existentially: a refutation gives a finite calculus proof; its expanded tree supplies a path; finite bounds and CRT supply the shared witnesses; the arithmetic graph existence theorems supply all auxiliary coordinates; and gates are evaluated. Positive representatives and positive carrier parameters are available as proved above. Consequently

$$\exists \boldsymbol{x} \in \mathbb{N}^{3614}\, P_{\mathrm{dir}}(\boldsymbol{x}) = 0 \quad \longleftrightarrow \quad \exists p\ \mathrm{Proof}_{\mathrm{ZFC}}(p, \bot).$$

The companion equation is linked by explicit gate projection and restoration. Taking negations proves the required equivalences.

9.2 What is inherited, and what has been established here

The mathematical dependencies are explicit:

1. The numerical code constructors and axiom/rule predicates from `Code254.lean` and `KCode222.lean`.
2. The repository statements `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. They connect this concrete calculus to the intended theory.
3. The Pell-index characterization and power graph, represented in Mathlib's `PellMatiyasevic` development and in the pinned repository Pell and power modules [3, 4].
4. For the integer family, the older native/compiler, signed-Pell, determinant-guard, exact-coefficient, and K215/refutation calibration of the 25-integer parent. Immediate sources are included in `coordinate_family/upstream`.

The new work proves the shared compiler theorem, supplies elementary graph arguments, explains the direct flow/carrier bridge, proves all new coordinate maps, and materializes and independently audits the actual finite polynomial data. It does not rebuild the transitive proof-assistant dependency closure of the inherited sources. It does not supply a new object-language ZFC proof term or claim that the Python programs establish logical soundness by themselves.

The proofs use finite induction, finite CRT, integer arithmetic and the cited number-theoretic theorems. These are provable in ZFC and apply to its internal natural numbers. There is no step requiring arithmetic soundness of ZFC, a model of ZFC, a standardness assumption about all proof codes, or Con(ZFC) as a hypothesis. The calibration theorems are mathematical dependencies to be checked on their merits, not a consistency assumption. This distinction is essential when expressing (1) as a theorem of ZFC.

9.3 Independent finite algebra checks

The direct checker uses a separate exact implementation to expand all 2,954 residual squares and compare the resulting coefficient dictionary with the delivered 15,996-term polynomial. It checks canonical indices, degree four, all variable occurrences, and the coefficient height. It also recovers all 2,569 eliminated affine definitions from the original rows and final variable names, without using the exporter's substitution map, and verifies the full restored row multiset.

The arithmetic controls include complete Pell certificates for six positive power examples, zero-exponent and zero-base controls for the optional total graph, changed-output rejection, 63 binomial cases including $k > n$, 20 factorial controls, 30 modulus-product controls, and a nontrivial shared CRT fixture. A static induction proves positivity at all 55 power call sites. These finite fixtures are regression checks; the universal validity comes from the mathematical graph theorems.

A separate compositional sparse-polynomial audit recognizes all 56 source wire definitions and all 34 branch residuals from the displayed numerical calculus. Branch fixtures and modular evaluations provide additional implementation controls. A separate symbolic power-graph audit substitutes all 61 definitional gates and matches all 17 remaining constraints to independently written versions of the cited Pell and power criteria, with exactly 27 additional natural witnesses.

For the integer family, another independently implemented parser and sparse polynomial engine imports none of the author's polynomial engine. It checks the exact row bridge from the literal incoming parent, all four triangular inverse identities, every retained and deleted row after each projection, variable liveness, degree bounds, and the three nonzero coefficient certificates. The full fixed tuple is also reconstructed from its finite recipe. Both audit families pass in ordinary Python and with assertions disabled using `python3-O`.

These checks do not search for a zero of the complete equations and do not claim that the equations have no zeros. Their purpose is to close the gap between the displayed algebra and the supplied literal artifacts.

10 Artifacts and offline reproduction

The archive is designed as a self-contained contribution directory:

`report/`
: This source, its included sections, the compiled PDF, and a shell build script. The article contains the ordinary proofs and states their scope.

`direct/`
: The two literal direct equations, fully expanded coefficient lists, deterministic generator, exact arithmetic and source audits, receipts, and immediate calibration/source provenance.

`coordinate_family/`
: The three new integer equations, their compact parent, exact six-numeral payloads and reconstruction recipe, proofs, inherited source material, comparison inventory, and independent checkers.

`verify.py`
: A root verification entry point. It checks packaged hashes and runs the independent algebra audits; its help lists optional reproduction checks.

`MANIFEST.sha256`
: Hashes of the packaged deliverables, excluding the manifest itself.

The numerical and polynomial audits use Python's standard library. Typical commands from the extracted directory are

```
python3 verify.py
python3 verify.py --help
cd report
sh build.sh
```

The document build uses a normal pdfLaTeX installation with the packages listed in the preamble. The JSON files are deliberately plain and inspectable; the mathematical definitions do not require trusting a particular executable serialization format. The hash manifest protects the identity of the files audited, not the truth of their mathematics.

11 What would constitute the next substantial advance?

The strongest conceptual gain here is reuse of one large arithmetic certificate across many local obligations. The factorial-free range step exposes several concrete remaining questions.

Share more than the global factorial. The thirteen coordinate bounds still require thirteen binomial graphs. The direct materialization contains 55 power certificates, so improvement of a shared multiple-binomial or shared-power graph would affect the largest repeated component. A correct construction must preserve each coordinate's decoded residue; simply combining vanishing products does not establish all coordinate bounds.

Replace the factorial scale by a cheaper sufficiently divisible scale. The shared theorem needs the relevant small differences and range factorials to divide T, and it needs every prime factor of each $1 + (i + 1)T$ to exceed the evaluation bounds. The factorial is a transparent way to guarantee all three. A smaller graph for a scale with these properties could help, but a numerical reduction in T alone need not reduce the number of polynomial unknowns.

Avoid duplicate evaluation circuits. The compiler evaluates a nonnegative majorant and a residue instance of the same large numerical rule circuit. A direct congruence circuit might share additional structure, but it must still provide an honest bound for prime decoding. Integer cancellation may make an expression smaller while destroying the nonnegative-coefficient monotonicity used in the proof.

Optimize the calibration rather than the simulated machine. A smaller directly calibrated proof calculus could reduce both circuit evaluations and the branch product. Such a proposal must include a complete equivalence with ZFC refutability. A short uncalibrated calculus, or an inconsistent strengthening, would not solve the stated problem.

Continue the integer coordinate search with degree-aware translations. The 21-variable construction succeeds because it keeps $Y = a - E$ and $k = \eta + \zeta$ as coordinates while eliminating other apparently equally natural choices. The first root is shifted before expansion; the second root is shifted before its companion coordinate is removed. A systematic search over integral triangular coordinates could uncover further cancellations. Exact zero-set inverses and a nonzero leading coefficient should be required for every candidate.

Measure several objectives together. The direct quartic with height 1,993 is larger than the variant with height 127,778,545. Both are meaningful outcomes of the same compiler. An optimization report should retain the full tuple of domain, unknowns, degree, height, monomials and source-circuit size. A change that is excellent for one objective may be poor for another.

These are specific avenues opened by the present constructions. None is being counted as a completed further improvement. Establishing an absolute minimum is a separate problem, and the deliverable makes no such claim.

References

[1] V. Reshetnikov, *ProveIt*, repository snapshot `08200eb10fde2db408c69dffd840d6920a569b1f`. https://github.com/VladimirReshetnikov/ProveIt. Relevant directories: `Computability/HilbertTenthProblem`, `Computability/DiophantineConsistency/con-zfc-diophantine-forms`, and `docs/incoming`.

[2] ProveIt, package 68, *Commutative certificates* and the *direct-flow-large-carrier* candidate, at the same snapshot. Immediate source text and the exact affine step are included with this contribution.

[3] The Mathlib community, *PellMatiyasevic*: Pell-index Diophantine representation and power characterization. https://leanprover-community.github.io/mathlib4_docs/Mathlib/NumberTheory/PellMatiyasevic.html. The relevant names include `Pell.matiyasevic` and `Pell.eq_pow_of_pell`. The immediate repository adaptations are `PellRepresentation.lean` and `PowerRepresentation.lean`.

[4] M. Carneiro, *A Lean formalization of Matiyasevič's theorem*, 2018. https://arxiv.org/abs/1802.01795.

[5] ProveIt incoming proposals at the pinned snapshot: *Joint Sign Guards*; *Structural Reductions, Round 11*; *Budget and Centering*; *Centered Pell Norms*; *Radical Guard Descent*; and `report.zip`. The complete archive-name/hash inventory and inspected root texts are included in the coordinate-family comparison and provenance files.

[6] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, 2026. https://arxiv.org/abs/2609.01594. This provides context for small-variable consistency equations; the constructions here are defined and proved through the dependencies explicitly stated above.